SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: Artificial Intelligence **COURSE CODE:** CSA17

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. *(BL3)*

Assessment Tool Weightage: 30%

QUESTIONS: 2

Duration : 45 Minutes
Total Marks:20

Annexure A

Descriptive Written Test

90/100

Code of Conduct:

I B. Priyanka *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Priyanka B
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Annexure A

Descriptive Written Test

1. A smart home automation system is being designed to manage lighting, temperature, and security based on user behavior and environmental conditions. The system must respond to real-time inputs such as motion detection, temperature changes, and time of day while also learning user preferences over time. In this context, explain the different types of intelligent agents (simple reflex, model-based, goal-based, and utility-based agents) and analyze how each type would function within this system. Justify which type of agent or hybrid approach would be most effective for ensuring comfort, energy efficiency, and security.

2. A robot is placed in an unknown maze and must find a path to the exit without prior knowledge of the layout. Explain how uninformed search strategies like Uniform Cost Search (UCS) and Iterative Deepening Search (IDS) can help solve this problem. Discuss the advantages and disadvantages of these algorithms in terms of time and space complexity. Relate the problem to different types of intelligent agents and explain how agent design affects decision-making. Suggest which combination of agent type and search strategy would be most effective and justify your choice.

CO1 - AT3

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Course code : CSA1737

Course name : Artificial Intelligence

(1) Intelligent Agents in a Smart Home Automation System

Introduction:

An intelligent agent is a system that perceives its environment through sensors and takes actions to achieve specific objectives.

Types of Intelligent Agents:

1. Simple Reflex Agent -

A simple reflex agent selects actions based only on the current input using predefined condition-action rules.

Eg: If motion is detected, turn on the lights

Advantages: Fast and simple to implement.

2. Model-Based Agent

A model-based agent maintains an internal state of the environment using current and previous observation

Eg: It remembers that a person entered a room and keeps the lights on even when motion is temporarily not detected.

Advantage: works partially effective.

3. Goal-Based Agent

A Goal-based agent selects actions to achieve a specific goal.

Eg: It adjusts the thermostat to maintain the desired room temperature while ensuring the house is locked at night.

Advantage: Make decisions based on desired outcomes.

4. Utility-Based Agent

A utility-based agent evaluates different actions based on their usefulness and selects the action that provides the overall outcome.

Eg: It balances comfort, electricity consumption and security.

Advantages: Balances multiple competing objectives.

Conclusion:

Thus, a hybrid intelligent agent provides fast responses, personalized control, reduced energy consumption and improved home security.

(2) Solving an Unknown maze using UCS and IDS

Introduction:

A robot placed in an unknown maze must explore its surroundings.

identify possible paths and reach the exit. Uninformed search algorithms explore states without using additional knowledge about which direction leads to the goal.

1. Uniform Cost Search (UCS)

Uniform cost search expands the node with the lowest total path cost from starting position.

advantages:

- Finds the least-cost path when movement costs are non-negative.

disadvantages:

- can consume large amounts of memory.

complexity: Time and space can be exponential in the worst case.

cost $\epsilon > 0$,

$O(b^{1+\lfloor c^*/\epsilon \rfloor})$ where c^* is the optimal solution cost.

2. Iterative Deepening Search (IDS)

Iterative deepening search repeatedly performs depth-limited search, increasing the depth limit after each unsuccessful attempt.

Advantages:

- requires less memory than UCS

Complexity:

Time: $O(b^d)$

space: $O(bd)$

b is the branching factor and d is the depth of the shallowest solution.

Conclusion:

IDS is appropriate for equal-cost movements and memory efficiency, UCS is better when finding the minimum-cost path is essential.